MOHAK SHARMA

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Product engineer and ML researcher with 3+ years shipping scalable systems - from fine-tuning LLMs on 60K+ job records to building real-time BCI pipelines in <100ms. Currently at Columbia (MS CS) and Subconscious AI as a Founding Engineer, driven by a deep interest in neuroscience, computational biology, and the science of human behavior. Fluent across full-stack engineering and machine learning, with a bias toward systems that are both research-rigorous and production-ready.

EDUCATION

Columbia University - Master of Science (Computer Science) — **New York, USA (Dec 2026)**
Guru Gobind Singh Indraprastha University - Bachelor of Technology (Computer Science) — **Delhi, India (2019 - 2023)**

PROJECTS & RESEARCH

CLARIS-Net: Neural-Driven Humanoid Avatar Control in Physics Simulation
- Developed an end-to-end BCI system translating noisy EEG motor imagery into real-time, physics-grounded humanoid actions with <100ms latency.
- Implemented a Deep RL (PPO/SAC) pipeline to refine decoded intent, generating natural bipedal gaits that adapt to user-specific neural signatures.
- Leveraged EEGNet and contrastive learning for robust signal decoding for >90% accuracy in continuous joint angle prediction for 17-DOF agents.

Neural Radiance Fields - 3D Reconstruction
- Engineered coarse-to-fine neural radiance field with stratified + importance sampling over 5000 training steps; optimized volume and neural rendering via sinusoidal positional encodings and learned view-dependent color decomposition with final loss of 0.005.
- Implemented full camera-to-ray rendering stack from intrinsics matrix construction and camera-to-world transforms to chunked inference, training at 2,048 rays per step (10.24 million rays total) with 4,096-ray render chunks; evaluated every 250 steps and auto-selected best checkpoint by PSNR.

MindSnap (Adaptive Intent Layer): Shared Autonomy for Multi-Modal Accessibility
- Designed a Smart Intent Layer using Partially Observable Markov Decision Processes (POMDPs) to model user goals as latent variables across modalities (screen, VR, game consoles), using Deep Q-Networks (DQN) for shared autonomy, providing proactive assistance (predictive snapping, haptic/auditory cuing) by maintaining a Bayesian belief state of user intent.

CausalBench
- Designed a multi-agent self-learning setup for causal evaluation that generates structural causal models and real-world counterfactual tasks, benchmarking GPT-4, Claude, and Gemini on confounding, selection bias, and counterfactual reasoning (40–65% accuracy on causal graph tasks).

Mechanistic Interpretability of AlphaFold2
- Applied UMAP-based dimensionality reduction and contrastive representation learning to 200K+ AlphaFold DB protein structures to model residue interactions, binding behavior, and geometric alignment across 48 transformer layers, revealing mechanistically interpretable substructures in AlphaFold2's representations, with implications for structure-based drug discovery and protein function understanding.

WORK EXPERIENCE

Founding Causal AI Engineer — **New York, USA**
Subconscious — **Apr 2026 - Present**
- Built an insights module surfacing AMCE estimates, WTP curves, and demographic heterogeneity from a Hierarchical Bayes conjoint model algorithms (Silverman KDE, AMCE aggregation) in a shared computation layer, enabling pricing, feature prioritization, and segment positioning decisions.
- Migrated a conjoint analysis platform from R to React + FastAPI, refactoring a 10,000-line monolith into a strict MVC architecture across 40+ modular components and reducing the root orchestrator to ~100 lines; introduced a Zustand-backed filter cache for reducing API costs by 3 times.

Forward Deployed Fullstack Engineer - Growth & Product — **Delhi, India**
Outscal — **Nov 2023 - Jul 2025**
- Architected an AI-powered job parsing pipeline processing 2,000+ jobs daily (60K+ total), fine-tuning Gemini on existing job data to extract structured information with 95%+ accuracy, demonstrating ability to build scalable ML systems that handle high-volume data processing in real time.
- Built custom GPT and Gemini models to generate learning content at scale - creating web stories, cheat sheets, video scripts, and quizzes - then developed a skill-matching recommendation system to dynamically link educational resources to relevant job listings.
- Designed and executed A/B experiments across user journey touchpoints, analyzing cohort behavior and funnel metrics in Posthog to test engagement strategies, driving 15% conversion lift through data-driven feature optimization and hypothesis-driven experimental methodology.
- Cut down key Web Vitals by up to 800% down to ~100 ms on critical pages by tuning page performance and SEO structured data.
- Led the end-to-end creation of an internal, no-code CMS and website builder platform, saving 60+ developer hours per week.

TECHNICAL SKILLS

- **Computer Vision & 3D:** 3D reconstruction, neural rendering, NeRFs, SfM/MVS pipelines (COLMAP, OpenMVG, Open3D), image-based rendering, OpenCV, PyTorch3D, multi-view geometry, camera calibration, pose estimation, 3D visualization (WebGL/Three.js-style viewers).
- **ML & Stats:** Python, PyTorch, TensorFlow, NumPy, pandas, scikit-learn, RNNs/LSTMs, transformers, CNNs, self-supervised learning, contrastive learning, causal inference, time-series modeling, probabilistic modeling.
- **BCI & Control:** EEG decoding, EEGNet, MuJoCo, PyBullet, real-time control systems, reinforcement learning (PPO, SAC, DDPG), continuous control, motor control, sensor fusion, POMDPs, Deep Q-Networks.
- **Scientific Computing:** UMAP, manifold learning, representation analysis, AlphaFold2, signal processing, spiking neuron models.
- **Systems & Tooling:** FastAPI, Node.js, Express, SQL, MongoDB, Redis, data pipelines, Linux, Git, AWS (EC2, S3, CloudWatch, Lambda).
- **Full-stack:** TypeScript, JavaScript, React, Next.js, Redux Toolkit, Zustand, Material UI, Tailwind, Node.js, Express, GraphQL.